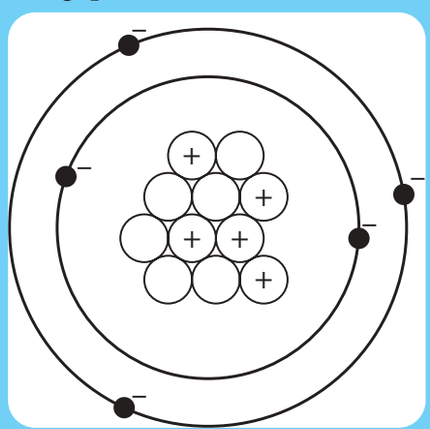
Table 4.1 Charge and mass of sub-atomic particles

Particle	Discoverer	Symbol	Charge	Mass (kg)
Proton	Goldstein	p	+1	1.6726×10^{-27}
Electron	Sir John Joseph Thomson	e	-1	9.1093×10^{-31}
Neutron	James Chadwick	n	0	1.6749×10^{-27}

ACTIVITY 2

Look at the given diagram and answer the following questions.



1. The positively charged particle is _____ .
2. The negatively charged particle is _____ .
3. _____ is the neutral particle.

4.3 Atomic number and Mass number

If all the elements are made up of same sub-atomic particles, how will a carbon atom differ from an iron atom? Further investigations led to the discovery that the number of protons inside the nucleus of an atom determines what element it is. For example, if the nucleus has only one proton, then all such atoms are hydrogen atoms. If there are eight protons then that atom is oxygen.

D8GZW

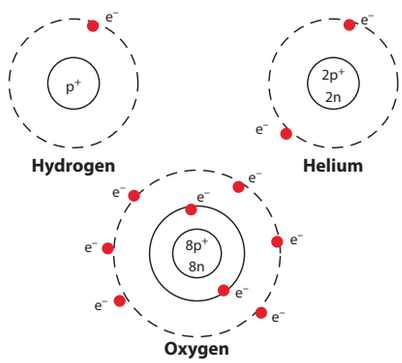
DO YOU KNOW?

Is the structure of an atom same as the structure of the solar system? Yes ! It is similar to the solar system. It has a core (center) called nucleus and it has paths called orbits around the nucleus.

4.3.1 Atomic number (z)

The number of electrons or protons in an atom is called the atomic number of that atom. It is represented by the letter Z. If we know the atomic number of an atom, we can find the number of electrons or protons in it.

Look at the figures. The nucleus of hydrogen atom has one proton around which revolves one electron. It means that its atomic number (z) is 1.



In a helium atom, there are two protons in the nucleus and two electrons revolving in the orbit around the nucleus. So, the atomic number(z) of helium is 2.

Look at the atomic structure of oxygen shown in the figure. What is its atomic number?

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